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Grade 32.25 out of 100.00

Question 1

Partially correct

Mark 2.75 out of 11.00

Recall the setting for reproducing kernel spaces from the lecture:

- Abstract data space X .
- A general kernel function $k : X \times X \rightarrow \mathbb{R}$.
- An associated reproducing kernel Hilbert space $\mathcal{H}$ with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and feature map $\Phi : X \rightarrow \mathcal{H}$.
- For $X = \mathbb{R}^n$ the Gauss kernel is defined as $k_{Gauss}(\mathbf{x}, \mathbf{y}) = \exp(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{\sigma^2})$ for some $\sigma > 0$.

Tick the correct statements related to this setting.

Select one or more:

- ☐ a. Kernel methods allow to compute $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ by just using k .
- ☐ b. In order to successfully work with kernels, you always need to compute an explicit formula for Φ .
- ☒ c. For the Gauss kernel, two points are regarded as similar, if their distance is at most σ . σ is a hyperparameter that has to be set beforehand. ✓
- ☒ d. Any symmetric function $f : X \times X \rightarrow \mathbb{R}$ so that $f \geq 0$ is a kernel function. ✗
- ☐ e. An equivalent formulation of k being a kernel function, is, that the associated Gram matrix $(k(\mathbf{x}_i, \mathbf{x}_j))_{i,j=1,\dots,l}$ for any $\mathbf{x}_1, \dots, \mathbf{x}_l \in X$ is symmetric and only has strictly negative eigenvalues.
- ☒ f. The reproducing property tells you that $f(\mathbf{x})$ is equal to $\langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}}$ for any $\mathbf{x} \in X$ and $f \in \mathcal{H}$. ✓
- ☐ g. The easier direction in Aronszajn's Theorem is to ensure the existence of $\mathcal{H}$ and Φ so that $\langle \Phi(\mathbf{x}), \Phi(\mathbf{y}) \rangle_{\mathcal{H}} = k(\mathbf{x}, \mathbf{y})$ for any $\mathbf{x}, \mathbf{y} \in X$ and a given kernel function k .
- ☐ h. To successfully apply kernel methods, you only need to perform linear methods in $\mathcal{H}$.

Question 2

Incorrect

Mark 0.00 out of 11.00

Let $w, x \in \mathbb{R}$. Let X be a random variable distributed according to a "supernice" parametrized distribution $p(x, w)$, so that p satisfies all the regularity conditions for any theorem we discussed in the lecture. X shall satisfy $E_p(X) = w^3 + w$ and by $I_p(w)$ we denote the associated Fisher information. Moreover, let the following estimators be given:

- Two estimators $\hat{w}_1$ and $\hat{w}_2$ of w so that $E_p(\hat{w}_1) = w$ and $E_p(\hat{w}_2) = 2w$ for any $w \in \mathbb{R}$.
- For an iid. sample of size l the associated Maximum Likelihood estimator (MLE) $\hat{w}_{ML}(l)$.

By $\mathcal{N}(\mu, \sigma^2)$ we, as usual, denote the Normal distribution with mean μ and variance σ^2 .

Tick the statements that are true for **any** choice of the terms defined above. In case there are statements about limits, the precise notion of convergence does not matter.

Select one or more:

- ☒ a. $\text{Var}(\hat{w}_1) \geq \frac{1}{I_p(w)}$ for any $w \in \mathbb{R}$. ✓
- ☐ b. $\text{Var}(\hat{w}_2) \leq I_p(w)$ for any $w \in \mathbb{R}$.
- ☐ c. $E_p(\hat{w}_{ML}(l)) = w$ for any $l \in \mathbb{N}$ and any $w \in \mathbb{R}$.
- ☒ d. In the limit $l \rightarrow \infty$, $\hat{w}_{ML}(l)$ converges to $\mathcal{N}(w, \frac{1}{I_p(w)})$. ✓
- ☒ e. For any $w \in \mathbb{R}$, $I_p(w)$ computes the average slope of the log-likelihood function associated to $p(\cdot, w)$. ✗
- ☐ f. The MLE for $E_p(X)$ for a sample of size l can be computed as $\hat{w}_{ML}^3(l) + \hat{w}_{ML}(l)$.
- ☒ g. The variances of $\hat{w}_1$, $\hat{w}_2$ and $\hat{w}_{ML}(l)$ can be bounded from below simultaneously by the Cramer Rao lower bound. ✗
- ☐ h. In the limit $l \rightarrow \infty$, $\hat{w}_{ML}(l)$ converges to w for any $w \in \mathbb{R}$.

Question 3

Correct

Mark 11.00 out of 11.00

A random sample X_1, X_2, X_3 of size 3 is drawn iid. from a population with mean $w = 3$ and variance $\sigma^2 = 4$.

Consider the estimators $\hat{w}_1 = \frac{1}{3}(X_1 + X_2 + X_3) + 1$ and $\hat{w}_2 = \frac{1}{4}(X_1 + 2X_2 + X_3) + 1$ of the mean w .

a) Which of the following statements about the estimators' biases is correct?

Both estimators are biased.



b) Compute the variance of the two estimators:

$\text{Var}(\hat{w}_1) =$

1.33



$\text{Var}(\hat{w}_2) =$

1.5



c) Based on the MSE, which estimator would you choose?

w_1



IMPORTANT: Only choose an answer if you are sure that it is correct. If you choose no answer, no points will be deducted. If you choose a wrong answer, it will cost you as many points as you would have gotten with the right answer.

Question 4

Partially correct

Mark 5.50 out of 11.00

Let us start recalling the general notion of a mixture model:

A mixture of K distributions $p(\mathbf{x}; \mathbf{w}_k)$ with $k = 1, \dots, K$ can be written as: $p(\mathbf{x}, \mathbf{W}, \pi) = \sum_{k=1}^K \pi_k p(\mathbf{x}; \mathbf{w}_k)$, where $\sum_{k=1}^K \pi_k = 1$ and all $0 \leq \pi_k \leq 1$ and $\mathbf{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_K\}$, $\pi = \{\pi_1, \dots, \pi_K\}$.

If the data are assumed to be drawn i.i.d. from $p(\mathbf{x}, \mathbf{W}, \pi)$, the likelihood function has the form

$$\mathcal{L}(\mathbf{X}, \mathbf{W}, \pi) = \ln p(\mathbf{X}, \mathbf{W}, \pi) = \ln \prod_{i=1}^l p(\mathbf{x}_i, \mathbf{W}, \pi) = \sum_{i=1}^l \ln \left(\sum_{k=1}^K \pi_k p(\mathbf{x}_i, \mathbf{w}_k) \right).$$

We also introduced the latent variables $\mathbf{u}_i$ as $p(u_{i,k} = 1) = \pi_k$ (for each $i = 1, \dots, l$ and $k = 1, \dots, K$) and their collection $\mathbf{U} = \{\mathbf{u}_1, \dots, \mathbf{u}_l\}$.

The so called responsibilities are then denoted by $\gamma(u_{i,j}) = \frac{\pi_j p(\mathbf{x}_i, \mathbf{w}_j)}{\sum_{k=1}^K \pi_k p(\mathbf{x}_i, \mathbf{w}_k)}$.

Tick the correct statements about the EM-algorithm, the approach which we used to maximize the likelihood function $\mathcal{L}(\mathbf{X}, \mathbf{W}, \pi)$.

Select one or more:

- ☒ a. Optimizing $E_{\mathbf{U}} \ln p(\mathbf{X}, \mathbf{U}, \mathbf{W}, \pi)$ instead of $\mathcal{L}(\mathbf{X}, \mathbf{W}, \pi)$ directly, allows to get rid of the sums inside of the logarithms and introduces the responsibilities. ✓
- ☒ b. The responsibilities account for the data distribution for a given cluster $j = 1, \dots, K$, i.e. $\gamma(u_{i,j}) = p(\mathbf{x}_i | j)$. ✗
- ☒ c. The responsibilities can be computed as an expectation of the latents $u_{i,k}$ over the posterior distribution $p(\mathbf{u}_i | \mathbf{x}_i, \mathbf{W}, \pi)$. ✓
- ☐ d. In the E-step, the responsibilities are kept fixed and the parameters $\mathbf{W}$ and π are optimized.
- ☒ e. The structure of the update rule for the coefficients π_k is analogous for any mixture model and can be derived via the method of Lagrange multipliers. ✓
- ☐ f. In the M-step, only the responsibilities are updated, where the current parameter values are kept fixed.
- ☒ g. $\mathbf{u}_i \in \mathbb{R}^K$ and it takes the value 1 only at one position, the other entries are 0. ✓
- ☒ h. For the distribution of the latents, we always have $p(\mathbf{U}) = \sum_{k=1}^K \pi_k^{u_k}$. ✗

Question 5

Partially correct

Mark 5.50 out of 11.00

Let us consider the standard statistical learning theory (SLT) setting from the lecture, i.e.

- Data points $\mathbf{Z} = (\mathbf{x}_i, y_i)_{i=1}^l$ sampled iid. from $p(\mathbf{x}, y)$ supported in $X \times \{-1, 1\}$
- A model class $\mathcal{G}$ of functions $g : X \rightarrow \{-1, 1\}$.
- The zero-one loss function L_{z0} .
- For $g \in \mathcal{G}$ and L_{z0} the associated generalization error (risk) $R(g)$ and the associated empirical error $R_{\text{emp}}(g, l)$.
- g_{opt} : the Bayes classifier, i.e. the model achieving the minimal risk. $g_{\mathcal{G}} = \arg \min_{g \in \mathcal{G}} R(g)$: the best model in the model class.
- g_l : an algorithm depending on the given data.

Tick the correct statements related to this setting. In case there are statements about limits, you do not need to care about the precise notion of convergence.

Select one or more:

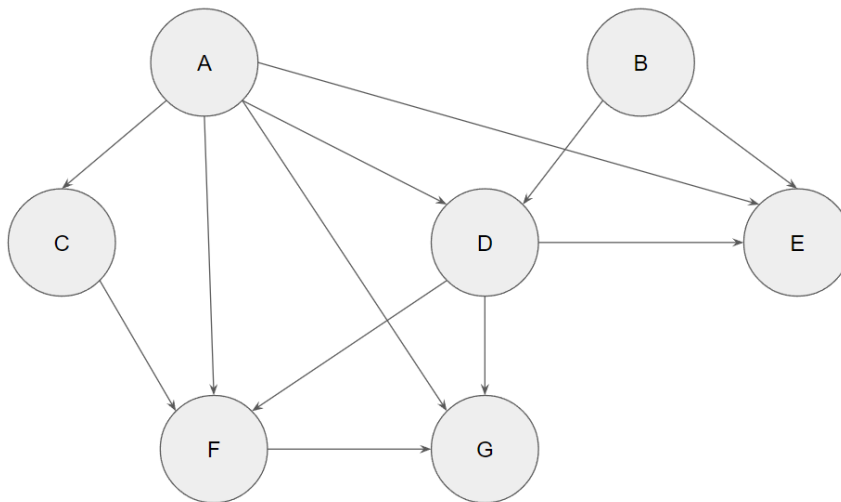
- ☒ a. For a fixed $g \in \mathcal{G}$, the convergence $R_{\text{emp}}(g, l)$ to $R(g)$ for $l \rightarrow \infty$ can be ensured by the law of large numbers. ✓
- ☒ b. For any $g \in \mathcal{G}$ and $\varepsilon > 0$, Hoeffding's inequality gives an upper bound on $p(|R_{\text{emp}}(g, l) - R(g)| \geq \varepsilon)$, that decays at most polynomially with respect to l . ✗
- ☐ c. The law of large numbers always ensures $R_{\text{emp}}(g_l, l) - R(g_l) \rightarrow 0$ for $l \rightarrow \infty$, since the variables $X_i = L(y_i, g_l(\mathbf{x}_i))$ for $i = 1, \dots, l$ are independent.
- ☒ d. The notion of consistency allows for statements about convergence of $R(g_l)$ to $R(g_{\text{opt}})$ or $R(g_{\mathcal{G}})$ for $l \rightarrow \infty$. ✓
- ☒ e. The approximation error is a deterministic statement about the "power" of $\mathcal{G}$. It tries to make statements about the "distance" of $\mathcal{G}$ from the Bayes classifier g_{opt} . ✓
- ☐ f. A crucial fact about SLT is, that you always assume some explicit knowledge about the data distribution in order to create your algorithms.
- ☐ g. The main purpose of the concept of uniform convergence is to ensure that $R(g)$ and $R_{\text{emp}}(g, l)$ are close for any $g \in \mathcal{G}$ and large l .
- ☐ h. Consistency always implies uniform convergence. The converse, however, is not true in general.

Question 6

Incorrect

Mark 0.00 out of 12.00

Given the following **Causal Graphical Model**:



Tick the boxes if the **corresponding statement is TRUE**.

For the Backdoor-criterion statements assume the following setup:

- The backdoor-criterion implies that for variables X, Y if we control for the right set of variables Z , the following holds:

$$P(Y|do(X)) = \sum_Z P(Y|X, Z)P(Z)$$
- Tick the "backdoor-criterion" boxes if with the provided sets X, Y, Z fulfill the backdoor criterion according to the formula above.

- ☐ a. The path $A - E - D - G$ is blocked
- ☒ b. The path $E - D - A - G$ is blocked ✖
- ☒ c. A is a mediator on the unblocked path $C - A - F - G$ ✖
- ☐ d. Controlling for G would make A and B statistically dependent
- ☒ e. The backdoor-criterion is fulfilled for $X = D, Y = E, Z = \{B\}$ ✖
- ☐ f. The backdoor-criterion is fulfilled for $X = C, Y = G, Z = \{A\}$
- ☒ g. The backdoor-criterion is fulfilled for $X = F, Y = G, Z = \{A, D\}$ ✔
- ☒ h. The backdoor-criterion is fulfilled for $X = D, Y = F, Z = \{A, G\}$ ✖

Question 7

Partially correct

Mark 5.67 out of 11.00

Consider students of a population who participate in an experiment, tracking their social media usage, sleep quantity and academic performance.

- The students spend M hours per day on social media.
- The students' sleep quantity is S hours per night.
- The students' academic performance rating is A (a score from 1 to 10).

Here are the Structural Equations with noise variables U_M , U_S , U_A that have been identified by fitting a model to the data.

$$M = 5 + U_M$$

$$S = 8 - 0.5 \cdot M + U_S$$

$$A = 4 - 0.4 \cdot M + 0.6 \cdot S + U_A$$

We are also given a specific student called "Forrest" with the following noise variables:

$$U_M = -1.8$$

$$U_S = -1.9$$

$$U_A = 0.1$$

a) The experiment's distribution as presented in the SCM is of the following type: ✓

b) Intervening on S would make M and A statistically independent: ✓

c) Based on the given noise variables for Forrest calculate M , S and A .

M : ✓

S : ✗

A : ✗

d) What would be the outcomes M' , S' and A' if Forrest had slept 8 hours?

M' : ✓

S' : ✓

A' : ✗

Question 8

Partially correct

Mark 1.83 out of 11.00

Consider a binary classification task in the 2-dimensional plane with labels $+1$ and -1 , respectively. Your model class $\mathcal{G}$ consists of all origin centered circles, i.e. for given radii $r > 0$, the model $g_r \in \mathcal{G}$ assigns the label $+1$ to all points which have distance less than r from the origin and -1 otherwise. By d we denote the corresponding VC-dimension of $\mathcal{G}$.

Tick the **correct** statements.

Hint: drawing some of the examples might help.

Select one or more:

- ☐ a. For the one point configuration consisting of $\mathbf{x}_1 = (0, 0)$ the label $y_1 = -1$ can not be realized. Thus we can conclude $d < 1$.
- ☒ b. For the two point configuration consisting of $\mathbf{x}_1 = (1, 0)$ and $\mathbf{x}_2 = (2, 0)$ the labelling $y_1 = +1$ and $y_2 = -1$ can be realized. ✗
Therefore $d \geq 2$.
- ☐ c. For the two point configuration consisting of $\mathbf{x}_1 = (1, 0)$ and $\mathbf{x}_2 = (0, 1)$, the labelling $y_1 = -1$ and $y_2 = +1$ can not be realized. Thus we can conclude $d < 2$.
- ☐ d. For any two points $\mathbf{x}_1$ and $\mathbf{x}_2$ denote their distances to the origin by r_1 and r_2 , respectively. Without loss of generality assume $r_1 \leq r_2$. Then the labelling $y_1 = -1$ and $y_2 = +1$ can not be realized, which ensures that $d < 2$.
- ☒ e. $d \geq 1$, since for the one point configuration consisting of $\mathbf{x}_1 = (1, 0)$ all possible labels can be realized. ✔
- ☒ f. $d = \infty$, since $\mathcal{G}$ consists of infinitely many models g_r . ✗
- ☐ g. For any two points $\mathbf{x}_1$ and $\mathbf{x}_2$ denote their distances to the origin by r_1 and r_2 , respectively. Without loss of generality assume $r_1 \leq r_2$. Then the labelling $y_1 = +1$ and $y_2 = -1$ can be realized, which ensures that $d \leq 2$.
- ☐ h. $d < \infty$, since $\mathcal{G}$ consists of only finitely many models g_r .

Question 9

Incorrect

Mark 0.00 out of 11.00

Tick the boxes if the corresponding statement is TRUE.

- ☐ a. To control for a set of variables X in Multiple Linear regression one has to add the set of variables X and an additional intercept to the set of independent variables.
- ☒ b. The identification-estimation procedure describes an abstract algorithm to execute causal inference. It starts with the causal estimate of interest, then uses causal model assumptions and some instance of the do-calculus to arrive at a statistical estimate. Given some data, the resulting estimand can be calculated. ✗
- ☒ c. The average causal effect of a treatment T on a drug D is defined as $ACE = \sum_D D \cdot P(D|do(T = 1)) - \sum_D D \cdot P(D|do(T = 0))$ ✓
- ☐ d. The do-calculus is complete (there are no further set of rules to work with the do-operator, that can't be derived from the do-calculus). The backdoor criterion is an instance (i.e. can be derived from) the do-calculus.
- ☒ e. Randomized Control Trials have the advantage of a very simple causal structure. The randomization part accounts for effects like placebo, whereas the control part accounts for effects like selection bias. ✗
- ☐ f. Given a Causal Graphical Model where A is a causal parent of B . An intervention on B would make B statistically independent of A .
- ☒ g. According to the ladder of causation, for a counterfactual to make sense, an intervention must have happened before. ✗
- ☒ h. Tick true if all of the following 3 statements are correct: (1) Correlation implies Association, (2) Causation implies Correlation, (3) Association does not imply Causation ✗